Indian Statistical Institute, Bangalore

B. Math (hons.) First Year

First Semester - Analysis I

Semestral Exam | Date: January 11, 2021

Maximum marks: 70 | Duration: 3 hours

1. Let $s_1 = 2$ and $s_{n+1} = \sqrt{6 + s_n}$ for al $n \geq 1$. Prove that $s_n \to 3$ (*Marks 15*).

2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function with no local maximum or local minimum. Prove that f is monotone (*Marks 15*).

3. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a differentiable function with derivative f'. Suppose $f'(x) = r$ if there is a sequence (x_n) such that $x_n \to x$ and $f'(x_n) = r$ for all n. Then f' is continuous at x(*Marks 15*).

4. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a differentiable function with bounded derivative. Suppose P is a polynomial such that $|P(x)| \leq |f(x)|$ for all $x \in \mathbb{R}$. Prove that degree of P is 0 or 1 (*Marks 10*).

5. Let $f\colon [a, b] \to \mathbb{R}$ be a differentiable function and there is a $M > 0$ such that $|f'(x)| \leq M|f(x) - f(a)|$ for all $x \in [a, b]$. Prove that $f' = 0$ (*Marks 15*).